Chapter 9

Complex Vectors and Matrices.

Real versus Complex

$\mathbf{R}$ = line of all real numbers $-\infty < x < \infty$ $\leftrightarrow$ $\mathbf{C}$ = plane of all complex numbers $z = x + iy$

$|x|$ = absolute value of x $\leftrightarrow$ $|\boldsymbol{z}| = \sqrt{\boldsymbol{x}^2 + \boldsymbol{y}^2} = \boldsymbol{r}$ = absolute value (or modulus) of z

1 and -1 solve $x^2 = 1$ $\leftrightarrow$ $z = 1, w, \ldots, w^{n-1}$ solve $z^n = 1$ where $w = e^{2\pi i/n}$

The **complex conjugate** of $z = x + iy$ is $\overline{\boldsymbol{z}} = \boldsymbol{x} - \boldsymbol{iy}$. $|z|^2 = x^2 + y^2 = z\overline{z}$ and $\dfrac{1}{z} = \dfrac{\overline{z}}{|z|^2}$.

The **polar form** of $z = x + iy$ is $|z|e^{i\theta} = \boldsymbol{re^{i\theta}} = \boldsymbol{r\cos\theta + ir\sin\theta}$. The angle has $\tan\theta = \dfrac{y}{x}$.

$\mathbf{R}^n$: vectors with n real components $\leftrightarrow$ $\mathbf{C}^n$: vectors with n complex components

length: $\|\boldsymbol{x}\|^2 = x_1^2 + \cdots + x_n^2$ $\leftrightarrow$ length: $\|\boldsymbol{z}\|^2 = |z_1|^2 + \cdots + |z_n|^2$

transpose: $(A^{\mathrm{T}})_{ij} = A_{ji}$ $\leftrightarrow$ conjugate transpose: $(A^{\mathrm{H}})_{ij} = \overline{A_{ji}}$

dot product: $\boldsymbol{x}^{\mathrm{T}}\boldsymbol{y} = x_1y_1 + \cdots + x_ny_n$ $\leftrightarrow$ inner product: $\boldsymbol{u}^{\mathrm{H}}\boldsymbol{v} = \overline{u}_1v_1 + \cdots + \overline{u}_nv_n$

reason for A^{T}: $(A\boldsymbol{x})^{\mathrm{T}}\boldsymbol{y} = \boldsymbol{x}^{\mathrm{T}}(A^{\mathrm{T}}\boldsymbol{y})$ $\leftrightarrow$ reason for A^{H}: $(A\boldsymbol{u})^{\mathrm{H}}\boldsymbol{v} = \boldsymbol{u}^{\mathrm{H}}(A^{\mathrm{H}}\boldsymbol{v})$

orthogonality: $\boldsymbol{x}^{\mathrm{T}}\boldsymbol{y} = 0$ $\leftrightarrow$ orthogonality: $\boldsymbol{u}^{\mathrm{H}}\boldsymbol{v} = 0$

symmetric matrices: $S = S^{\mathrm{T}}$ $\leftrightarrow$ Hermitian matrices: $S = S^{\mathrm{H}}$

$\boldsymbol{S} = \boldsymbol{Q\Lambda Q^{-1}} = \boldsymbol{Q\Lambda Q}^{\mathrm{T}}$(real Λ) $\leftrightarrow$ $\boldsymbol{S} = \boldsymbol{U\Lambda U^{-1}} = \boldsymbol{U\Lambda U}^{\mathrm{H}}$ (real Λ)

skew-symmetric matrices: $K^{\mathrm{T}} = -K$ $\leftrightarrow$ skew-Hermitian matrices $K^{\mathrm{H}} = -K$

orthogonal matrices: $Q^{\mathrm{T}} = Q^{-1}$ $\leftrightarrow$ unitary matrices: $U^{\mathrm{H}} = U^{-1}$

orthonormal columns: $Q^{\mathrm{T}}Q = I$ $\leftrightarrow$ orthonormal columns: $U^{\mathrm{H}}U = I$

$(Q\boldsymbol{x})^{\mathrm{T}}(Q\boldsymbol{y}) = \boldsymbol{x}^{\mathrm{T}}\boldsymbol{y}$ and $\|Q\boldsymbol{x}\| = \|\boldsymbol{x}\|$ $\leftrightarrow$ $(U\boldsymbol{x})^{\mathrm{H}}(U\boldsymbol{y}) = \boldsymbol{x}^{\mathrm{H}}\boldsymbol{y}$ and $\|U\boldsymbol{z}\| = \|\boldsymbol{z}\|$

Sec 9.1

Review: Complex numbers

Complex numbers.

- Complex numbers extend the concept of the reals through the definition of an imaginary number:

$$i \equiv \sqrt{-1}$$

- Complex numbers have a real part and an imaginary part, and can be visualized as points on the complex plane:

$$z = a + bi \ \rightarrow z = (a, b)$$

- The absolute value (magnitude) of a complex number is the distance of the point from the origin in the complex plane:

$$|z| = \sqrt{a^2 + b^2}$$

- The complex conjugate simplifies this definition:

$$|z|^2 = \bar{z}z, \qquad \bar{z} = a - bi$$

$\mathbf{R}$ = line of all real numbers $-\infty < x < \infty \quad \leftrightarrow \quad \mathbf{C}$ = plane of all complex numbers $z = x + iy$

$|x|$ = absolute value of $x \quad \leftrightarrow \quad |\boldsymbol{z}| = \sqrt{\boldsymbol{x^2 + y^2}} = \boldsymbol{r}$ = absolute value (or modulus) of z

1 and -1 solve $x^2 = 1 \quad \leftrightarrow \quad z = 1, w, \ldots, w^{n-1}$ solve $z^n = 1$ where $w = e^{2\pi i/n}$

The **complex conjugate** of $z = x + iy$ is $\overline{\boldsymbol{z}} = \boldsymbol{x - iy}$. $|z|^2 = x^2 + y^2 = z\overline{z}$ and $\dfrac{1}{z} = \dfrac{\overline{z}}{|z|^2}$.

The **polar form** of $z = x + iy$ is $|z|e^{i\theta} = \boldsymbol{re^{i\theta}} = \boldsymbol{r\cos\theta + ir\sin\theta}$. The angle has $\tan\theta = \dfrac{y}{x}$.

Sec 9.2

Hermitian and Unitary Matrices

Conjugate transpose

Definition: When you transpose of a complex vector or matrix, you need to take the complex conjugate too.

Conjugate transpose $\quad \overline{\boldsymbol{z}}^{\mathrm{T}} = \begin{bmatrix} \overline{z}_1 & \cdots & \overline{z}_n \end{bmatrix} = \begin{bmatrix} a_1 - ib_1 & \cdots & a_n - ib_n \end{bmatrix}.$

Definition: Instead of a bar for the conjugate and T for the transpose, we use a new superscript H to define the conjugate transpose of z.

$\boldsymbol{A}^{\mathrm{H}}$ **is "*A* Hermitian"** $\quad$ If $\quad A = \begin{bmatrix} 1 & i \\ 0 & 1+i \end{bmatrix} \quad$ then $\quad A^{\mathrm{H}} = \begin{bmatrix} 1 & 0 \\ -i & 1-i \end{bmatrix}$

Dot products and complex numbers

- For vectors of real values, we defined a dot product as:

$$\vec{u} \cdot \vec{v} = \vec{u}^T \vec{v} = \sum_{i=1}^{n} u_i v_i$$

- Dot product is closely linked to the concept of length/magnitude:

$$\|\vec{u}\|^2 = \vec{u}^T \vec{u} = \sum_{i=1}^{n} |u_i|^2$$

- For complex numbers, magnitude involves the complex conjugate:

$$\|\vec{z}\|^2 \equiv \sum_{i=1}^{n} |z_i|^2 = \sum_{i=1}^{n} \bar{z}_i z_i = \vec{z}^H \vec{z}$$

Inner product of complex vectors

- For vectors of complex values, we define the dot product as:

$$\vec{u} \cdot \vec{v} = \vec{u}^H \vec{v} = \sum_{i=1}^{n} \overline{u}_i v_i$$

- Note that the order matters!

$$\vec{u}^H \vec{v} = \sum_{i=1}^{n} \overline{u}_i v_i \neq \sum_{i=1}^{n} \overline{v}_i u_i = \vec{v}^H \vec{u}$$

- Instead the complex dot product is conjugate symmetric:

$$\vec{u}^H \vec{v} = (\vec{v}^H \vec{u})^H \rightarrow \vec{u} \cdot \vec{v} = \overline{\vec{v} \cdot \vec{u}}$$

- Orthogonality and orthonormality are based on the same definitions:

$$\vec{u}^H \vec{v} = 0, \qquad \vec{u}^H \vec{u} = 1$$

Example:

Find the inner product of $u = \begin{bmatrix} 1 \\ i \end{bmatrix}$ $and\ v = \begin{bmatrix} i \\ 1 \end{bmatrix}$.

The inner product of $\boldsymbol{u} = \begin{bmatrix} 1 \\ i \end{bmatrix}$ with $\boldsymbol{v} = \begin{bmatrix} i \\ 1 \end{bmatrix}$ is $\begin{bmatrix} 1 & -i \end{bmatrix} \begin{bmatrix} i \\ 1 \end{bmatrix} = 0.$

DEFINITION The inner product of real or complex vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ is $\boldsymbol{u}^{\mathrm{H}}\boldsymbol{v}$:

$$\boldsymbol{u}^{\mathrm{H}}\boldsymbol{v} = \begin{bmatrix} \overline{u}_1 & \cdots & \overline{u}_n \end{bmatrix} \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} = \overline{u}_1 v_1 + \cdots + \overline{u}_n v_n.$$

Example:

1. Find the length of (1, i) and the squared length (1+i, 1-i).

2. What is the vector that has length 0?

Now the squared length of $(1, i)$ is $1^2 + |i|^2 = 2$. The length is $\sqrt{2}$. The squared length of $(1 + i, 1 - i)$ is 4. The only vectors with zero length are zero vectors.

Conjugate transpose of matrices

The inner product of Au with v equals the inner product of u with $A^{\mathrm{H}}v$:

$$(Au)^{\mathrm{H}}v = u^{\mathrm{H}}(A^{\mathrm{H}}v).$$

A^{H} is also called the "adjoint" of A

The conjugate transpose of AB is $(AB)^{\mathrm{H}} = B^{\mathrm{H}}A^{\mathrm{H}}.$

Example:

Find the Hermitian matrix for A:

$$A = \begin{bmatrix} 1 & i \\ 0 & 1+i \end{bmatrix}$$

$\boldsymbol{A}^{\mathrm{H}}$ **is "*A* Hermitian"** If $A = \begin{bmatrix} 1 & i \\ 0 & 1+i \end{bmatrix}$ then $A^{\mathrm{H}} = \begin{bmatrix} 1 & 0 \\ -i & 1-i \end{bmatrix}$

Theorems for symmetric complex matrices

If $S = S^{\mathrm{H}}$ and z is any real or complex column vector, the number $z^{\mathrm{H}}Sz$ is real.

Quick proof: $z^{\mathrm{H}}Sz$ is certainly 1 by 1. Take its conjugate transpose:

$$(z^{\mathrm{H}}Sz)^{\mathrm{H}} = z^{\mathrm{H}}S^{\mathrm{H}}(z^{\mathrm{H}})^{\mathrm{H}} \quad \text{which is } z^{\mathrm{H}}Sz \text{ again.}$$

So the number $z^{\mathrm{H}}Sz$ equals its conjugate and must be real.

Theorems for symmetric complex matrices

Every eigenvalue of a Hermitian matrix is real.

Proof Suppose $Sz = \lambda z$. *Multiply both sides by* z^{H} *to get* $z^{\mathrm{H}}Sz = \lambda z^{\mathrm{H}}z$. On the left side, $z^{\mathrm{H}}Sz$ is real. On the right side, $z^{\mathrm{H}}z$ is the length squared, real and positive. So the ratio $\lambda = z^{\mathrm{H}}Sz/z^{\mathrm{H}}z$ is a real number. Q.E.D.

Theorems for symmetric complex matrices

The eigenvectors of a Hermitian matrix are orthogonal (when they correspond to different eigenvalues). If $S\boldsymbol{z} = \lambda\boldsymbol{z}$ and $S\boldsymbol{y} = \beta\boldsymbol{y}$ then $\boldsymbol{y}^{\mathrm{H}}\boldsymbol{z} = 0$.

Proof Multiply $S\boldsymbol{z} = \lambda\boldsymbol{z}$ on the left by $\boldsymbol{y}^{\mathrm{H}}$. Multiply $\boldsymbol{y}^{\mathrm{H}}S^{\mathrm{H}} = \beta\boldsymbol{y}^{\mathrm{H}}$ on the right by $\boldsymbol{z}$:

$$\boldsymbol{y}^{\mathrm{H}}S\boldsymbol{z} = \lambda\boldsymbol{y}^{\mathrm{H}}\boldsymbol{z} \quad \text{and} \quad \boldsymbol{y}^{\mathrm{H}}S^{\mathrm{H}}\boldsymbol{z} = \beta\boldsymbol{y}^{\mathrm{H}}\boldsymbol{z}. \tag{5}$$

The left sides are equal so $\lambda\boldsymbol{y}^{\mathrm{H}}\boldsymbol{z} = \beta\boldsymbol{y}^{\mathrm{H}}\boldsymbol{z}$. Then $\boldsymbol{y}^{\mathrm{H}}\boldsymbol{z}$ **must be zero**.

The inner product of complex function space

- A commonly used inner product for function space is the integral of the product of the two functions over the functions' domain:

$$\langle f(x), g(x) \rangle = \int_a^b f(x)g(x)dx$$

- If we have complex functions, we make the same sort of change:

$$\vec{u}^T\vec{v} = \sum_{i=1}^{n} u_i v_i \rightarrow \vec{u}^H\vec{v} = \sum_{i=1}^{n} \overline{u}_i v_i$$

- Take the complex conjugate of the second function!

$$\langle f(x), g(x) \rangle = \int_a^b f(x)\overline{g(x)}dx$$

  - Note, this domain of integration implies that the input argument is real.

Projections in complex vector space:

- **Recall that the projection of a real vector onto another vector or onto a vector space is given by:**

$$proj_{\vec{v}}(\vec{u}) = \frac{\vec{u}^T\vec{v}}{\vec{v}^T\vec{v}}\vec{v}, \qquad proj_V(\vec{u}) = A(A^TA)^{-1}A^T\vec{u}, A = [\vec{v}_i]$$

- **For complex vectors, we simply use the conjugate transpose:**

$$proj_{\vec{v}}(\vec{u}) = \frac{\vec{u}^H\vec{v}}{\vec{v}^H\vec{v}}\vec{v}, \qquad proj_V(\vec{u}) = A(A^HA)^{-1}A^H\vec{u}, A = [\vec{v}_i]$$

- **Gram-Schmidt is based on projections, so the general rule becomes:**

$$\vec{w}_i = \vec{a}_i - \sum_{j=1}^{i-1} \frac{\vec{a}_i^H\vec{w}_j}{\vec{w}_j^H\vec{w}_j}\vec{w}_j$$

Unitary matrices

Definition: A unitary matric Q is a complex square matrix that has orthonormal columns.

Every matrix Q with orthonormal columns has $Q^{\mathrm{H}}Q = I$.

If Q is square, it is a unitary matrix. Then $Q^{\mathrm{H}} = Q^{-1}$.

If Q is unitary then $\|Qz\| = \|z\|$. *Therefore* $Qz = \lambda z$ *leads to* $|\lambda| = 1$.

The EVD and SVD with complex matrices.

- **The complex extension of an orthogonal matrix is called a unitary matrix:**

$$Q^T = Q^{-1} \rightarrow Q^H = Q^{-1}$$

  - **All columns in a unitary matrix are orthonormal.**

- **The complex extension of a symmetric matrix is called a Hermitian matrix:**

$$S^T = S \rightarrow S^H = S$$

- **A Hermitian matrix is always diagonalizable with orthogonal eigenvectors and has only REAL eigenvalues.:**

$$S = S^H \rightarrow S = Q\Lambda Q^H,\ Q^H = Q^{-1},\ \lambda_i \in \mathbb{R},\ \forall i$$

- **All matrices have a well-defined Singular Value Decomposition:**

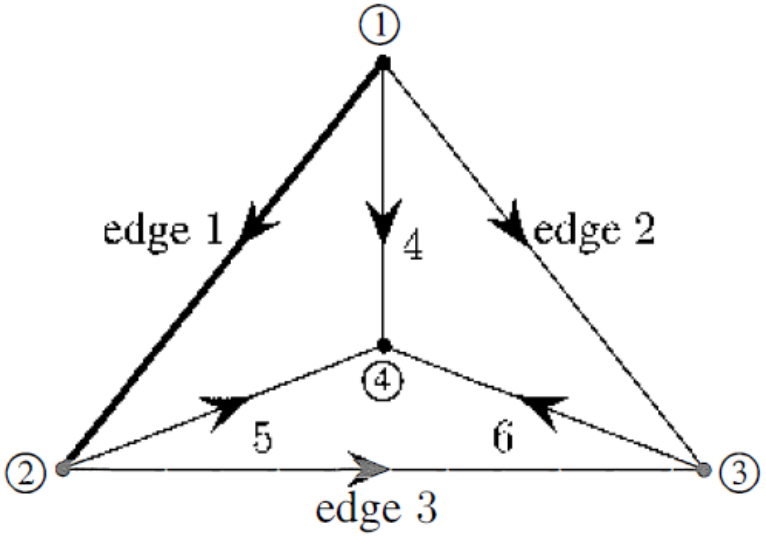
$$A = U\Sigma V^H,\ U^H = U^{-1},\ V^H = V^{-1},\ \sigma_i \geq 0 \in \mathbb{R}, \forall i$$

$$A^H A = S = Q\Lambda Q^H,\ Q = V,\ \Lambda = \Sigma^H\Sigma \rightarrow AV = U\Sigma$$

Sec 9.3 and beyond

Applications of Linear Algebra

Incidence Matrix

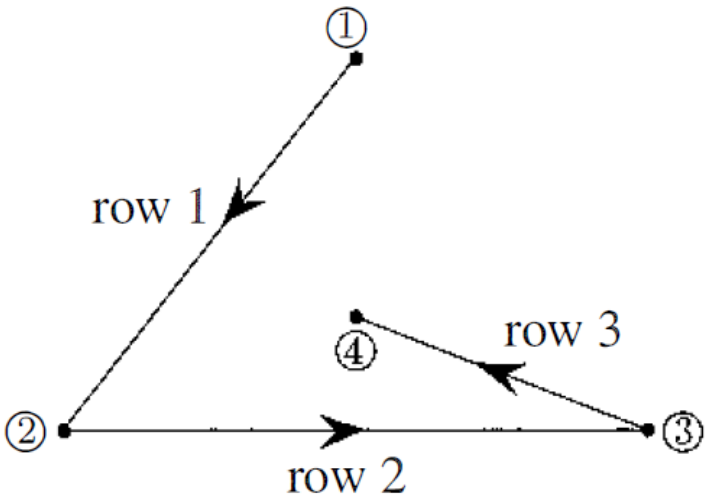


$$
\begin{array}{cc}
 & \text{node} \\
 & \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\
A = & \begin{bmatrix} -1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{array} \text{ edge}
\end{array}
$$

Figure 10.1: Complete graph with $m = 6$ edges and $n = 4$ nodes: 6 by 4 incidence matrix A.

Figure 10.1 displays a graph with $m = 6$ edges and $n = 4$ nodes. The 6 by 4 matrix A tells which nodes are connected by which edges. The first row $-1, 1, 0, 0$ shows that the first edge goes *from node* 1 *to node* 2 (-1 for node 1 because the arrow goes out, $+1$ for node 2 with arrow in).

Example: Find the matrix from the graph



$$B = \begin{array}{c} \begin{array}{cccc} ① & ② & ③ & ④ \end{array} \\ \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix} \end{array} \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \quad \text{edge}$$

Graph theory and Linear Algebra

Observation 1: The first graph is complete-every pair of the nodes is connected by an edge.

Observation 2: The second graph is a tree-the graph has no closed loops.

Question: Could we connect these two graphs by Linear Algebra?

Graph theory vs Linear Algebra

- Elimination reduces graphs to a tree.
  - Loops produce dependent rows in A and zero rows in row echelon form U and R.
  - Example: Edges 1, 2, 3 form a loop in the first graph.

$$\begin{bmatrix} -1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 1 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & -1 & 1 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix}$$

Applications in Physics

The incidence matrix A is a difference matrix.

When x_1, x_2, x_3, x_4 are voltages at the nodes, Ax gives voltage differences:

$$Ax = \begin{bmatrix} -1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} x_2 - x_1 \\ x_3 - x_1 \\ x_3 - x_2 \\ x_4 - x_1 \\ x_4 - x_2 \\ x_4 - x_3 \end{bmatrix}.$$

The null space of Ax=0 and Bx=0

- Method 1: Directly solve by algebra. All six voltage differences are zero. Then all four voltages are equal. So, the null space is x= (c, c, c, c). It works for Bx = 0 as well.

- Method 2: Use Gaussian Elimination to solve. You will find three pivots in both A and B. Then there exists one free variable. So, the dimension of the null space is 1.

The row space

- Method 1: Linear combination of the independent rows (rows of B).
- Method 2: Orthogonality. V is in the row space if and only if it is perpendicular to (1,1,1,1) in the null space.

The column space

- Column space: Linear combination of independent columns.
  - Question: How many independent columns do we have here?
    - It must be three since there are three independent rows.
  - Question 2: How can we tell if a particular b is in the column space of an incidence matrix?
    - Solve Ax =b directly.
    - Use Kirchhoff's voltage law: The component of Ax =b add to zero around every loop.

The Left Null space

The left null space is generated by the solutions to $A^T y = 0$.

The dimension is = 6-3=3.

Current Law

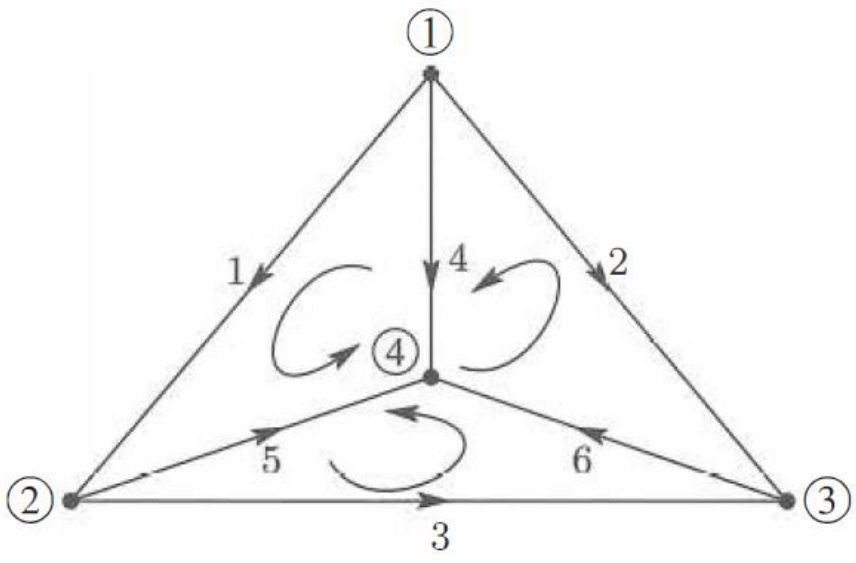
$$A^{\mathrm{T}}\boldsymbol{y} = \begin{bmatrix} -1 & -1 & 0 & -1 & 0 & 0 \\ 1 & 0 & -1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \\ y_6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

What do the equations mean? The first equation says that $-y_1 - y_2 - y_4 = 0$. ***The net flow into node 1 is zero***. The fourth equation says that $y_4 + y_5 + y_6 = 0$. ***Flow into node 4 minus flow out is zero***. The equations $A^{\mathrm{T}}\boldsymbol{y} = \mathbf{0}$ are famous and fundamental:

Kirchhoff's Current Law: $A^{\mathrm{T}}\boldsymbol{y} = 0$ ***Flow in equals flow out at each node.***

Summary

A unit of current goes around the big triangle, forward on edges 1 and 3, and backward on edge 2, the six currents are y=(1,-1, 1, 0, 0, 0)



$$\begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ -1 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \\ 0 \\ 1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

3 small loops big loop

The incidence matrix A comes from a connected graph with n nodes and m edges. The row space and column space have dimensions $r = n - 1$. The nullspaces of A and A^{T} have dimensions 1 and $m - n + 1$:

$\boldsymbol{N}(A)$ The constant vectors $(c, c, \ldots, c)$ make up the nullspace of A : dim $= 1$.

$\boldsymbol{C}(A^{\mathrm{T}})$ The edges of any tree give r independent rows of A : $r = n - 1$.

$\boldsymbol{C}(A)$ ***Voltage Law***: The components of $A\boldsymbol{x}$ add to zero around all loops: dim $= n - 1$.

$\boldsymbol{N}(A^{\mathrm{T}})$ ***Current Law***: $A^{\mathrm{T}}\boldsymbol{y} =$ **(flow in)** $-$ **(flow out)** $= \mathbf{0}$ is solved by loop currents.

There are $m - r = m - n + 1$ independent small loops in the graph.

Euler formula vs linear algebra

For every graph in a plane, linear algebra yields ***Euler's formula***: Theorem 1 in topology!

$$(\textit{number of nodes}) - (\textit{number of edges}) + (\textit{number of small loops}) = \mathbf{1}.$$

$$\text{This is } (n) - (m) + (m - n + 1) = 1.$$

Application: Linear Programming

The matrix has $n > m$, more unknowns than equations. If there are any solutions $\boldsymbol{x} \geq \mathbf{0}$ to $A\boldsymbol{x} = \boldsymbol{b}$, there are probably a lot. Linear programming picks the solution $\boldsymbol{x}^* \geq \mathbf{0}$ that minimizes the cost:

> ***The cost is $c_1x_1 + \cdots + c_nx_n$. The winning vector x^* is the nonnegative solution of $Ax = b$ that has smallest cost.***

Thus a linear programming problem starts with a matrix A and two vectors $\boldsymbol{b}$ and $\boldsymbol{c}$:

i) A has $n > m$: for example $A = \begin{bmatrix} 1 & 1 & 2 \end{bmatrix}$ (one equation, three unknowns)

ii) $\boldsymbol{b}$ has m components for m equations $A\boldsymbol{x} = \boldsymbol{b}$: for example $\boldsymbol{b} = \begin{bmatrix} 4 \end{bmatrix}$

iii) The ***cost vector*** $\boldsymbol{c}$ has n components: for example $\boldsymbol{c} = \begin{bmatrix} 5 & 3 & 8 \end{bmatrix}$.

Then the problem is to minimize $\boldsymbol{c} \cdot \boldsymbol{x}$ subject to the requirements $A\boldsymbol{x} = \boldsymbol{b}$ and $\boldsymbol{x} \geq 0$:

Minimize $5x_1 + 3x_2 + 8x_3$ ***subject to*** $x_1 + x_2 + 2x_3 = 4$ ***and*** $x_1, x_2, x_3 \geq 0$.

Feasible Solution sets

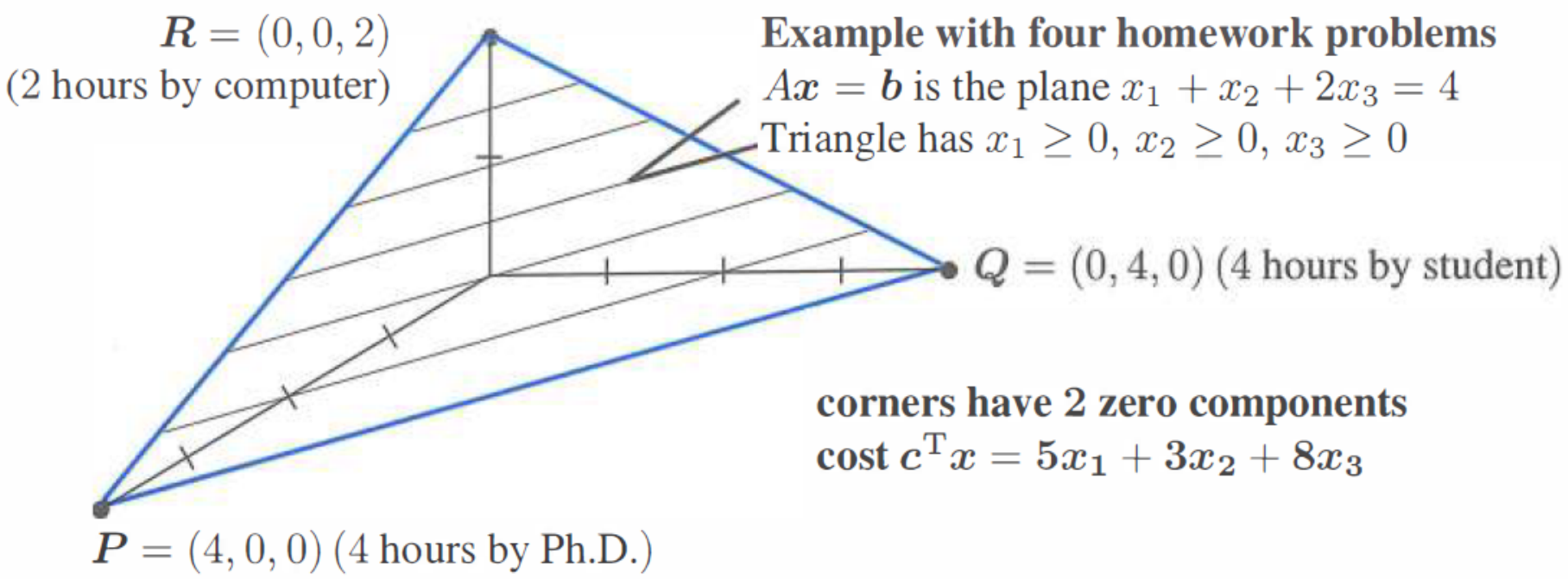


Figure 10.5: The triangle contains all nonnegative solutions: $Ax = b$ and $x \geq 0$. The lowest cost solution x^* is a corner P, Q, or R of this feasible set.

Best solution

The first plane $5x_1 + 3x_2 + 8x_3 = C$ to touch the triangle has minimum cost C. ***The point where it touches is the solution x^*.*** This touching point must be one of the corners $\boldsymbol{P}$ or $\boldsymbol{Q}$ or $\boldsymbol{R}$. A moving plane could not reach the inside of the triangle before it touches a corner! So check the cost $5x_1 + 3x_2 + 8x_3$ at each corner:

$\boldsymbol{P} = (4, 0, 0)$ costs 20 $\quad$ $\boldsymbol{Q} = (0, 4, 0)$ costs 12 $\quad$ $\boldsymbol{R} = (0, 0, 2)$ costs 16.

The winner is $\boldsymbol{Q}$. Then $\boldsymbol{x}^* = (0, 4, 0)$ solves the linear programming problem.

Notes

- Some linear programming programs maximize profit instead of minimizing cost, but the math is the same.
- The requirement or constraints could be impossible to satisfy.
- It could also happen the feasible set is unbounded and the minimum could be $-\infty$ for this case.
- Linear programming problems come in pairs, the dual problem.

The dual problem

- Original problem: Find the minimum cost $C^T x$ subject to the constraint Ax =b.
- The dual problem: transposes A and switches b and c.
  - Maximize $b \cdot y$ subject to $A^T y \leq c$.

$$\textit{If } Ax = b, x \geq 0, A^{\mathrm{T}} y \leq c \textit{ then } b^{\mathrm{T}} y = (Ax)^{\mathrm{T}} y = x^{\mathrm{T}}(A^{\mathrm{T}} y) \leq x^{\mathrm{T}} c.$$

If either problem has a best vector (x^* or y^*) then so does the other.

Minimum cost $c \cdot x^*$ equals maximum income $b \cdot y^*$